

AI Architecture Papers Portal — Independent Test Plan

Application: AI Architecture Papers Portal
Version: 1.0
Date: 2026-06-12
Server URL: http://localhost:5000
Stack: Flask (Python) backend, Vanilla HTML/CSS/JS frontend, SQLite database
Prepared for: Independent Site Testers

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1. Test Environment & Prerequisites

Setup

Item	Detail
Server	Flask dev server on http://localhost:5000
Python	3.14 (global install)
Database	SQLite (papers.db) — auto-created on first run
Paper folder	AI papers for WebProject1/ (must contain at least 5 PDFs with JSON sidecars)
Browsers to test	Chrome (latest), Edge (latest), Firefox (latest), Safari (latest on macOS if available)

Network	Corporate proxy configured; Azure OpenAI endpoint reachable (for AI tests)
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Starting the Server

```
cd WebProject1
start.bat
```

Server should start on port 5000 and print the watched folder path.

Test Data Checklist

Before testing, confirm:

- [] At least 5 PDFs exist in `AI_papers_for_WebProject1/`
- [] At least 3 PDFs have `.json` sidecar files with full metadata
- [] At least 2 PDFs have no sidecar (for auto-inference tests)
- [] At least 1 PDF has a `.jpg/.jpeg` infographic image alongside it
- [] At least 1 sidecar has an explicit `"groups"` array (admin-pinned)
- [] `papers.db` has been deleted or is fresh (for first-user bootstrap test)

2. Authentication & Authorization

TC-AUTH-001: First User Bootstrap (Auto-Registration)

Step	Action	Expected Result
1	Ensure <code>papers.db</code> has no users (delete DB or verify <code>user_count = 0</code>)	—
2	Navigate to <code>http://localhost:5000/</code>	Redirected to <code>/login</code>
3	Enter a username and password, click "Sign In"	Account is auto-created; redirected to main page
4	Log out and log back in with the same credentials	Login succeeds

TC-AUTH-002: Login Page — Valid Credentials

Step	Action	Expected Result
1	Navigate to <code>/login</code>	Login form is displayed with Username, Password fields, and "Sign In" button
2	Enter valid username and password	Redirected to <code>/</code> (main page)
3	Check page title	"Login — AI Architecture Papers Portal" on login page

TC-AUTH-003: Login Page — Invalid Credentials

Step	Action	Expected Result
1	Enter incorrect username or password on <code>/login</code>	Redirected to <code>/login?error=Invalid+username+or+password</code>

2	Verify error message is displayed	Red error alert reads "Invalid username or password"
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TC-AUTH-004: Registration Page

Step	Action	Expected Result
1	Navigate to <code>/register</code>	Registration form with Username, Password, Confirm Password fields
2	Submit with valid data (password $\geq$ 4 chars, passwords match)	Account created; auto-logged in; redirected to <code>/</code>
3	Submit with mismatched passwords	Error: "Passwords do not match"
4	Submit with password < 4 chars	Error: "Password must be at least 4 characters"
5	Submit with empty username	Error: "Username and password are required"
6	Submit with already-taken username	Error: "Username already taken"

TC-AUTH-005: Registration Page Links

Step	Action	Expected Result
1	On <code>/register</code> , click "Sign in" link	Navigates to <code>/login</code>
2	On <code>/login</code> , click "Create one" link	Navigates to <code>/register</code>

TC-AUTH-006: Logout

Step	Action	Expected Result
1	While logged in, navigate to <code>/logout</code>	Session ends; redirected to <code>/login</code>
2	Try to access <code>/</code> directly	Redirected to <code>/login</code>

TC-AUTH-007: Unauthenticated API Access

Step	Action	Expected Result
1	Without logging in, call GET <code>/api/papers</code>	Returns 401 JSON: <code>{"error": "Authentication required"}</code>
2	Without logging in, call GET <code>/api/visit-count</code>	Returns 401
3	Without logging in, navigate to <code>/</code>	Redirected to <code>/login?next=/<code></code></code>

TC-AUTH-008: Authenticated Redirect After Login

Step	Action	Expected Result
1	While unauthenticated, navigate to <code>/</code>	Redirected to <code>/login?next=/<code></code></code>
2	Log in with valid credentials	Redirected back to <code>/</code> (the original target)

TC-AUTH-009: Already Authenticated — Login/Register Pages

Step	Action	Expected Result
1	While logged in, navigate to <code>/login</code>	Redirected to <code>/</code> (main page)
2	While logged in, navigate to <code>/register</code>	Redirected to <code>/</code> (main page)

3. Main Page Load & Layout

TC-LOAD-001: Initial Page Load

Step	Action	Expected Result
1	Log in and load <code>/</code>	Page displays hero header, filter strip, two-column layout, and footer
2	Check hero section	Contains eyebrow text "SYSTEMS ARCHITECTURE AND INNOVATION", title, description
3	Verify filter buttons	Four buttons visible: All, Most Important, Most Read, Latest
4	"All" button is active by default	Has <code>is-active</code> class with dark background

TC-LOAD-002: Visit Counter

Step	Action	Expected Result
1	Load the main page	Visit count in DB increments by 1
2	Call <code>GET /api/visit-count</code>	Returns JSON <code>{"count": N}</code> where $N \geq 1$
3	Reload the page	Visit count increments again

TC-LOAD-003: Static File Serving

Step	Action	Expected Result
1	Load <code>/styles.css</code>	CSS file returned with correct content type
2	Load <code>/app.js</code>	JavaScript file returned
3	Load a PDF: <code>/AI papers for WebProject1/{filename}.pdf</code>	PDF file downloads/displays
4	Load a keyword image: <code>/Key Word Images/{name}.png</code>	Image displays

TC-LOAD-004: Two-Column Desktop Layout

Step	Action	Expected Result
1	View on screen $> 1100\text{px}$ wide	Two-column grid: feed panel (left, $\sim 320\text{-}390\text{px}$), detail panel (right, fills remaining)

2	Both panels are sticky	Panels remain visible while scrolling the full summaries section
3	Feed panel contains: Discovery section, Paper Feed, Summary Search	All three sub-panels visible and scrollable

4. Paper Feed & Filtering

TC-FEED-001: Paper Feed Population

Step	Action	Expected Result
1	Load main page	Paper feed shows cards for all PDFs in the local library
2	Verify card structure	Each card shows: thumbnail image, group tag(s), title, preview text, authors, year
3	Check feed count badge	Shows "N papers" matching the actual count
4	Feed title shows "Paper Feed" or "Paper Feed — All"	Correct

TC-FEED-002: Filter — All

Step	Action	Expected Result
1	Click "All" filter button	All papers appear in the feed
2	Feed title updates	Shows "Paper Feed — All"
3	Button has <code>is-active</code> state	Dark background, white text

TC-FEED-003: Filter — Most Important

Step	Action	Expected Result
1	Click "Most Important" filter button	Exactly 5 papers appear (auto-scored, excluding admin-pinned)
2	Feed title updates	Shows "Paper Feed — Most Important"
3	Feed count badge	Shows "5 papers"
4	Open browser console	Log entry <code>[Importance] Top 5:</code> visible with paper titles and scores

TC-FEED-004: Filter — Most Read

Step	Action	Expected Result
1	Click "Most Read" filter button	Only papers with "read" group appear
2	These include papers whose filename contains "survey", "technical report", or "benchmark"	Correct matches shown
3	Feed count updates	Matches visible card count

TC-FEED-005: Filter — Latest

Step	Action	Expected Result
1	Click "Latest" filter button	All papers appear (every paper gets "latest" group)
2	Feed title updates	Shows "Paper Feed — Latest"

TC-FEED-006: Filter Button Toggle Behavior

Step	Action	Expected Result
1	Click "Most Important" then click "Most Read"	Only "Most Read" is active; previous deactivated
2	Active filter button has visual differentiation	is-active class with distinct styling
3	Clicking same filter twice	Filter remains active (does not deselect)

TC-FEED-007: Most Important Tooltip

Step	Action	Expected Result
1	Hover over "Most Important" button	Tooltip appears explaining the scoring formula
2	Tooltip content	Mentions Tier 1/Tier 2 keywords, age term, top 5, admin-pinned exemption
3	Move mouse away	Tooltip disappears (160ms fade)

TC-FEED-008: Paper Card Selection

Step	Action	Expected Result
1	Click any paper card in the feed	Card gets is-selected class (border + shadow highlight)
2	Previously selected card	Loses is-selected class
3	Detail panel updates	Shows the clicked paper's details

TC-FEED-009: Sidecar vs Auto-Inferred Data

Step	Action	Expected Result
1	Select a paper WITH a sidecar JSON	Title, authors, year, summary match the sidecar data
2	Select a paper WITHOUT a sidecar JSON	Title is auto-generated from filename; year is regex-extracted or defaults to 2024

TC-FEED-010: Admin-Pinned Papers (Sidecar Groups Override)

Step	Action	Expected Result
1	Verify a paper with explicit "groups" in its sidecar JSON	Paper uses the sidecar-defined groups, not auto-scored

2	This paper's pinned flag is 1 in the DB	Confirmed via GET /api/papers/{id}
3	Paper is excluded from importance auto-scoring	Does not appear in "Most Important" unless "important" is in its sidecar groups

5. Paper Detail Panel

TC-DETAIL-001: Detail Panel Rendering

Step	Action	Expected Result
1	Click a local paper card	Detail panel shows: keyword images, title, authors/year, groups, summary, datacenter significance, key result signal
2	Action links for local paper	"Jump to Full Section" link + "Open Local PDF" link

TC-DETAIL-002: Keyword Image Trio

Step	Action	Expected Result
1	View detail panel for any paper	Three keyword images displayed in a flex row
2	Each has a figcaption label	Keywords derived from paper title/preview/summary
3	Images are 100% width × 180px height with object-fit: cover	Correctly sized

TC-DETAIL-003: Jump to Full Section

Step	Action	Expected Result
1	Click "Jump to Full Section" link	Page scrolls to the corresponding #paper-{id} anchor in the Full Summaries section

TC-DETAIL-004: Open Local PDF

Step	Action	Expected Result
1	Click "Open Local PDF" link	PDF opens in a new tab or downloads

TC-DETAIL-005: Discovery Paper in Detail Panel

Step	Action	Expected Result
1	Click a discovery card	Detail panel shows "Web Discovery" badge instead of "Jump to Full Section"
2	Link shows "Open Discovery Source"	Opens the external source URL in a new tab

6. Full Paper Summaries Section

TC-SUMMARY-001: Full Summary Rendering

Step	Action	Expected Result
1	Scroll below the two-column layout	Full Paper Summaries section visible
2	Each paper has an <code><article></code> with anchor ID <code>paper-{id}</code>	Correct
3	Each summary includes: title, keyword images, authors, year, category, summary text, datacenter significance, key result signal, PDF link	All fields rendered

TC-SUMMARY-002: Summary Section Count

Step	Action	Expected Result
1	Count summary sections	Matches total number of local library papers

7. Summary Search

TC-SEARCH-001: Basic Search

Step	Action	Expected Result
1	Type a keyword in the summary search input	Matching summary sections remain visible; non-matching sections get hidden
2	Match count updates	Shows "N matches" in <code>#summarySearchCount</code>
3	Matching text is highlighted	<code><mark class="summary-search-highlight"></code> wraps matches (green background #d6f5e3)

TC-SEARCH-002: No Matches

Step	Action	Expected Result
1	Type a query that matches no papers	All sections hidden; count shows "No matches"

TC-SEARCH-003: Clear Search

Step	Action	Expected Result
1	Type a query, then clear the input (backspace or clear button)	All sections reappear; highlights removed
2	Press Escape key while search input is focused	Input is cleared and blurred

TC-SEARCH-004: Safe DOM Highlighting

Step	Action	Expected Result
1	Search for <code><script>alert(1)</script></code>	No script execution; treated as literal text
2	Search for HTML entities	No DOM injection; TreeWalker-based highlighting is safe

8. Web Discovery Panel

TC-DISC-001: Discovery Panel Structure

Step	Action	Expected Result
1	Verify discovery panel location	Top of feed panel, inside a <code><details open></code> element
2	Title row shows "Web Discovery"	With result count badge
3	"Find New Papers" button is visible	Clickable
4	Status text shows "Ready" initially	<code>aria-live="polite"</code> attribute present

TC-DISC-002: Collapse/Expand Discovery Panel

Step	Action	Expected Result
1	Click the <code><summary></code> row	Discovery panel collapses; caret changes to ▀
2	Click again	Panel expands; caret changes to ▴

TC-DISC-003: Find New Papers

Step	Action	Expected Result
1	Click "Find New Papers" button	Status changes to "Searching..."
2	Wait for results	Discovery feed populates with cards; status shows "Updated — N results (M already in your library)"
3	Results appear from OpenAlex and/or arXiv	At least some results shown (network permitting)

TC-DISC-004: Find New Papers Tooltip

Step	Action	Expected Result
1	Hover over "Find New Papers" button	Tooltip explains it searches OpenAlex + arXiv and results stay in the staging area

TC-DISC-005: Discovery Card Structure

Step	Action	Expected Result
1	View a discovery card	Shows: image, "Discovery" tag, title, preview, authors/year

2	Footer has "Open Source" link	Opens external URL in new tab
3	Footer has "Download Metadata" button	Present with tooltip on hover

TC-DISC-006: Discovery Card Selection

Step	Action	Expected Result
1	Click a discovery card body	Card gets <code>is-selected</code> class (accent border + box-shadow)
2	Detail panel updates to show discovery paper	Includes "Web Discovery" badge
3	Click "Open Source" link in card footer	Link opens; card selection does NOT change (stopPropagation)

TC-DISC-007: Download Metadata

Step	Action	Expected Result
1	Click "Download Metadata" on a discovery card	A <code>.json</code> file downloads
2	Inspect the downloaded JSON	Contains: <code>title</code> , <code>authors</code> , <code>year</code> , <code>preview</code> , <code>summary</code> , <code>datacenter</code> , <code>metrics</code> , <code>link</code>
3	JSON is in sidecar format	Compatible with placing next to a PDF for library import

TC-DISC-008: Cross-Deduplication

Step	Action	Expected Result
1	Run discovery when a local paper matches a discovered title	Matching discovery results are removed from the feed

TC-DISC-009: Self-Deduplication

Step	Action	Expected Result
1	Observe discovery results	No duplicate titles within the discovery feed

TC-DISC-010: Discovery Hard Cap

Step	Action	Expected Result
1	Run discovery multiple times	No more than 24 discovery results retained

TC-DISC-011: Rotating Query Pool

Step	Action	Expected Result
1	Click "Find New Papers" multiple times	Each click uses the next query from the 16-query pool
2	After 16 clicks	Query pool wraps around to the first query

9. Discovery Search & Ranking

TC-DSEARCH-001: Discovery Search API

Step	Action	Expected Result
1	Call GET <code>/api/discover/search?q=transformer&year_fsource_counts;dtc=2026</code>	Returns JSON with results, query, <code>_fsource_counts</code> ; <code>dtc=2026</code>
2	Verify <code>source_counts</code>	Contains keys: <code>core-pr</code> , <code>openalex</code> , <code>arxiv</code> with integer values
3	Empty query	Returns empty results with <code>empty_reason</code> message

TC-DSEARCH-002: Discovery Rank API

Step	Action	Expected Result
1	Call POST <code>/api/discover/rank</code> with team, query, candidates	Returns ranked papers with architecture scores
2	Missing team parameter	Returns 400 with error message listing valid teams
3	Valid teams	<code>oie</code> , <code>e2o</code> , <code>ai_on_ia</code> , <code>hickory_delta</code>

TC-DSEARCH-003: Ranking Criteria API — GET

Step	Action	Expected Result
1	Call GET <code>/api/discover/ranking-criteria?team=oie</code>	Returns criteria with key, question, slider, weight, weight_percent
2	Five criteria keys	<code>compute_arch_fit</code> , <code>memory_hierarchy_impact</code> , <code>cluster_scalability</code> , <code>implementation_readiness</code> , <code>efficiency_tco</code>
3	Missing/invalid team	Returns 400 error

TC-DSEARCH-004: Ranking Criteria API — POST (Update)

Step	Action	Expected Result
1	POST new slider values and questions for a team	Updated criteria returned
2	Slider values clamped	Values stay within <code>0.0</code> to <code>10.0</code>
3	<code>reset_defaults: true</code>	Criteria reset to defaults for that team

TC-DSEARCH-005: Discovery Progress API

Step	Action	Expected Result
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1	Call GET <code>/api/discover/progress</code>	Returns JSON with stage, active, processed, total, found, source_counts, message
2	Before any search	Stage is "idle", message is "Ready"

10. Image System

TC-IMG-001: Keyword Image Resolution Chain

Step	Action	Expected Result
1	Select a paper whose keywords match a local PNG in Key Word Images/	Local PNG displayed
2	Select a paper whose keywords have no local PNG	Wikipedia thumbnail fetched and displayed
3	If Wikipedia fetch fails	SVG placeholder shown (300×200, #eef2f8 background, keyword text)

TC-IMG-002: PDF.js Thumbnails

Step	Action	Expected Result
1	View a local paper card	PDF page 1 thumbnail renders as the card image
2	Thumbnail dimensions	Render width ~480px, JPEG quality 0.82
3	Subsequent views of same paper	Cached thumbnail used (no re-render)

TC-IMG-003: Infographic Image

Step	Action	Expected Result
1	Select a paper that has a .jpg alongside its PDF	Infographic image displayed in the detail panel

TC-IMG-004: Generated Images Folder

Step	Action	Expected Result
1	Check AI papers for <code>WebProject1/generated/</code>	Directory exists (auto-created at startup)
2	After generating images via API	Files saved here as <code>{paper_id}_figure.jpg</code> or <code>{paper_id}_infographic.jpg</code>

11. Paper CRUD API

TC-API-001: List All Papers

Step	Action	Expected Result
1	GET /api/papers	Returns JSON array of all local papers
2	Each paper object	Contains: id, filename, title, authors, year, groups, preview, summary, datacenter, metrics, link, isLocal: true

TC-API-002: Search Papers

Step	Action	Expected Result
1	GET /api/papers?q=transformer	Returns papers whose metadata matches the query
2	Empty query	Returns all papers

TC-API-003: Get Single Paper

Step	Action	Expected Result
1	GET /api/papers/{valid_id}	Returns single paper JSON
2	GET /api/papers/nonexistent-id	Returns 404

TC-API-004: Create Paper (Upload PDF)

Step	Action	Expected Result
1	POST /api/papers with multipart form: pdf file + optional title, authors, year, preview, summary, datacenter, metrics	Returns 201 with created paper JSON
2	Verify PDF saved	File exists in AI papers for WebProject1/ with secure_filename() applied
3	SSE event fired	Connected browsers receive "reload" event
4	Upload non-PDF file	Returns 400: "File must be a PDF"
5	Upload without file	Returns 400: "No PDF file provided"
6	Upload duplicate filename	Returns 409: "A paper with this filename already exists"

TC-API-005: Update Paper Metadata

Step	Action	Expected Result
1	PUT /api/papers/{id} with JSON body {"title": "New Title"}	Returns updated paper JSON
2	Nonexistent paper ID	Returns 404
3	Empty/invalid fields	Returns 400: "No valid fields provided"

TC-API-006: Delete Paper

Step	Action	Expected Result
1	DELETE /api/papers/{id}	Returns 204 No Content
2	PDF file removed from disk	Confirmed
3	JSON sidecar removed from disk (if it existed)	Confirmed
4	SSE event fired	Connected browsers receive "reload" event
5	Delete nonexistent paper	Returns 404

TC-API-007: Upload Size Limit

Step	Action	Expected Result
1	Upload a file > 100 MB	Returns 413 Request Entity Too Large

TC-API-008: Citation Counts API

Step	Action	Expected Result
1	GET /api/papers/citation-counts	Returns JSON object with {paper_id: count_or_null} for all papers
2	Counts fetched from OpenAlex	Logged in server console

12. AI-Powered Features

Note: These tests require Azure OpenAI to be configured in `azure_openai_config.txt`. If AI is unavailable, verify graceful degradation.

TC-AI-001: Generate Infographic

Step	Action	Expected Result
1	POST /api/papers/{id}/generate-infographic	Returns JSON with generated_infographic path or errors
2	Image saved to generated/ folder	File exists as JPEG
3	Paper record updated	generated_infographic field set in DB
4	Nonexistent paper ID	Returns 404

TC-AI-002: Generate Images (Best Figure + Infographic)

Step	Action	Expected Result
1	POST /api/papers/{id}/generate-images	Returns JSON with best_figure and generated_infographic paths
2	Best figure extracted from PDF via GPT-4o vision	JPEG file saved in generated/
3	Both fields updated in DB	Confirmed via GET

TC-AI-003: Discovery Figure Extraction

Step	Action	Expected Result
1	POST /api/discover/figure with {"pdf_url": "https://arxiv.org/pdf/2401.12345"}	Returns status, figure_base64, page, bbox
2	Status "found"	Base64 image data present
3	Status "none"	Reason indicates text-only or no pages
4	No pdf_url	Returns 400

TC-AI-004: Manual Bounding Box

Step	Action	Expected Result
1	POST /api/discover/figure/manual-bbox with pdf_url, manual_bbox (normalized 0..1)	Returns ok: true, bbox_validated: true
2	Bbox validation	x1 < x2, y1 < y2, min 5% width/height
3	Invalid bbox	Returns 400 with error
4	Feedback saved to figure_feedback table	Confirmed

TC-AI-005: PDF Page Render

Step	Action	Expected Result
1	POST /api/discover/pdf-page with {"pdf_url": "...", "page": 0}	Returns image_base64, page, page_count, base_bbox, width, height
2	DPI capped between 96 and 300	Confirmed
3	Missing pdf_url	Returns 400

TC-AI-006: AI Unavailable Graceful Degradation

Step	Action	Expected Result
1	Remove or invalidate azure_openai_config.txt	—
2	Call any AI endpoint	Returns appropriate error (e.g., 503 for report, 502/error status for figure)
3	Non-AI features continue working	Paper feed, filtering, search, etc. all functional

13. Clipboard & Reporting

TC-CLIP-001: Download ZIP

Step	Action	Expected Result
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1	POST /api/clipboard/download-zip with { <code>"papers": [{pdf_url, title, id}, ...]</code> }	Returns ZIP file (application/zip)
2	ZIP contains PDFs named after slugified titles	Confirmed
3	Papers with missing <code>pdf_url</code>	ZIP entry is a <code>_error.txt</code> file instead
4	Empty papers array	Returns 400

TC-CLIP-002: Generate Report

Step	Action	Expected Result
1	POST /api/clipboard/report with { <code>"papers": [...], "mode": "summary"</code> }	Returns JSON with <code>report</code> text (one-page summary)
2	Empty papers list	Returns 400
3	AI not configured	Returns 503 with descriptive error

14. Server-Sent Events (SSE) — Live Updates

TC-SSE-001: SSE Connection

Step	Action	Expected Result
1	Open browser on main page	EventSource connects to /api/changes
2	Check Network tab	SSE stream open with <code>text/event-stream</code> content type
3	Wait 25+ seconds without changes	Receive heartbeat : <code>ping</code> comment

TC-SSE-002: Live Paper Reload

Step	Action	Expected Result
1	Open the site in two browser tabs	Both connected via SSE
2	Drop a new PDF into AI papers for WebProject1/	Within 5 seconds, both tabs receive "reload" event
3	Paper feed updates in both tabs	New paper appears without manual refresh

TC-SSE-003: SSE on Paper CRUD

Step	Action	Expected Result
1	Create a paper via API (POST /api/papers)	SSE "reload" fires; open browsers update
2	Update a paper via API (PUT /api/papers/{id})	SSE "reload" fires
3	Delete a paper via API (DELETE /api/papers/{id})	SSE "reload" fires; paper disappears from feed

TC-SSE-004: SSE Requires Authentication

Step	Action	Expected Result
1	Without logging in, try to connect to /api/changes	Returns 401 (not a valid SSE stream)

15. Responsive Design & Cross-Browser

TC-RESP-001: Desktop (> 1100px)

Step	Action	Expected Result
1	Browser width > 1100px	Two-column grid: feed panel (~320-390px) + detail panel
2	Grid gap is 18px	Verified in DevTools
3	Feed card images are 128px wide	Verified
4	Max container width is 1200px, centered	Verified

TC-RESP-002: Small Laptop (960px–1100px)

Step	Action	Expected Result
1	Resize to ~1000px width	Grid columns shrink to minmax(280px, 340px) 1fr
2	Grid gap reduces to 14px	Verified
3	Card images shrink to 100px	Verified

TC-RESP-003: Mobile (< 960px)

Step	Action	Expected Result
1	Resize to < 960px	Single-column layout (1fr)
2	Feed panel	position: relative, height: auto (not sticky)
3	Detail panel	position: relative, max-height: none, overflow-y: visible
4	Card images shrink to 104px	Verified
5	Discovery feed max-height	280px

TC-RESP-004: Narrow (< 600px)

Step	Action	Expected Result
1	Resize to < 600px	Tooltips capped at max-width: calc(100vw - 32px)

2	"Most Important" tooltip	Left-aligned (no centering transform), caret at left: 24px
3	No horizontal scrolling	Confirmed

TC-RESP-005: Cross-Browser Scrollbar Stability

Step	Action	Expected Result
1	Test in Edge	scrollbar-gutter: stable prevents layout shift when scrollbar appears/disappears
2	Test in Chrome	Same behavior
3	Test in Firefox	Same behavior

TC-RESP-006: Dynamic Header Offset

Step	Action	Expected Result
1	Load page at various zoom levels (100%, 125%, 150%)	--header-offset CSS variable recalculated correctly
2	Resize browser window	--header-offset updates on resize event
3	Sticky panels fill remaining viewport height	No overflow or gap

16. Accessibility

TC-A11Y-001: Keyboard Navigation

Step	Action	Expected Result
1	Tab through filter buttons	All four buttons receive focus
2	Press Enter/Space on filter button	Filter activates
3	Tab through paper feed cards	Cards (buttons) receive focus
4	Press Enter/Space on focused card	Card is selected; detail panel updates

TC-A11Y-002: ARIA Attributes

Step	Action	Expected Result
1	Filter strip	role="toolbar" with aria-label="Paper category filter"
2	Paper cards	role="option", aria-selected="true/false", aria-label="Select {title}"
3	Paper feed container	role="listbox"
4	Discovery status	aria-live="polite"
5	Summary search count	aria-live="polite"

TC-A11Y-003: Focus Visibility

Step	Action	Expected Result
1	Tab to any interactive element	Visible focus ring via <code>:focus-visible</code>
2	Test in Chrome, Edge, Firefox	Focus ring consistent

TC-A11Y-004: Alt Text

Step	Action	Expected Result
1	Inspect keyword images	<code>alt</code> attribute present with keyword name
2	Inspect card images	<code>alt</code> attribute present with paper title

TC-A11Y-005: Reduced Motion

Step	Action	Expected Result
1	Enable "Reduce motion" in OS accessibility settings	All CSS transitions and animations disabled
2	Verify in DevTools	<code>@media (prefers-reduced-motion: reduce)</code> rules active

TC-A11Y-006: Color Contrast

Step	Action	Expected Result
1	Test primary text (#132028) on page background (#f4f6f2)	WCAG AA compliant (contrast ratio $\geq 4.5:1$)
2	Test tag badge text (white) on badge backgrounds (#9e2f2f, #205ea6, #1f6b54)	WCAG AA compliant
3	Test muted text (#4a5d66) on panel background (#fcdfdc)	WCAG AA compliant

17. Performance & Stability

TC-PERF-001: Initial Load Time

Step	Action	Expected Result
1	Clear cache, load main page	Page renders with paper feed within 3 seconds (on localhost)
2	Check Network tab	No requests exceeding 5 seconds

TC-PERF-002: Concurrent SSE Connections

Step	Action	Expected Result
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1	Open the site in 5 browser tabs	All 5 maintain SSE connections
2	Drop a new PDF into the paper folder	All 5 tabs receive the reload event
3	Server remains responsive	API calls still return quickly

TC-PERF-003: Large Library (50+ Papers)

Step	Action	Expected Result
1	Add 50+ PDFs to the paper folder	Feed loads all papers without significant lag
2	Filtering still responsive	Sub-second UI updates
3	Summary search still responsive	Highlighting works without visible delay

TC-PERF-004: Threaded Server

Step	Action	Expected Result
1	While SSE stream is active, make API calls	API calls succeed (not blocked by SSE)
2	Multiple simultaneous API requests	All return correctly (threaded mode)

TC-PERF-005: Folder Watcher Stability

Step	Action	Expected Result
1	Leave server running for 30+ minutes	Watcher thread continues polling every 5 seconds
2	Add/remove PDFs during this time	Changes detected and reported
3	No memory leaks or thread crashes	Server console shows no errors

18. Security

TC-SEC-001: CSRF Protection — Login Form

Step	Action	Expected Result
1	Verify login form uses <code>method="POST"</code>	Confirmed
2	Verify registration form uses <code>method="POST"</code>	Confirmed

TC-SEC-002: Password Storage

Step	Action	Expected Result
1	Create a user account	—
2	Inspect <code>papers.db</code> → <code>users</code> table	<code>password_hash</code> column contains a Werkzeug hash, NOT plaintext

TC-SEC-003: XSS Prevention — Summary Search

Step	Action	Expected Result
1	Search for <code></code>	Treated as literal text; no script execution
2	DOM-based highlighting via TreeWalker	No innerHTML injection

TC-SEC-004: XSS Prevention — Error Messages

Step	Action	Expected Result
1	Navigate to <code>/login?error=<script>alert(1)</script></code>	Error message rendered as text, not executed
2	Error display uses <code>textContent</code> , not <code>innerHTML</code>	Confirmed in source

TC-SEC-005: Filename Sanitization

Step	Action	Expected Result
1	Upload a PDF with path traversal name: <code>../../../../etc/passwd.pdf</code>	<code>secure_filename()</code> sanitizes to safe name
2	File saved only within <code>AI papers for WebProject1/</code>	No path traversal

TC-SEC-006: Authentication on All API Endpoints

Step	Action	Expected Result
1	Without login, call each API endpoint (papers, discover, clipboard, SSE)	All return <code>401</code> or redirect to login
2	Verify no endpoint is publicly accessible	Confirmed

TC-SEC-007: Upload Size Enforcement

Step	Action	Expected Result
1	Attempt to upload file > 100 MB	Rejected by Flask's <code>MAX_CONTENT_LENGTH</code> (100 MB)

19. Error Handling & Edge Cases

TC-ERR-001: Empty Paper Folder

Step	Action	Expected Result
1	Remove all PDFs from <code>AI papers for WebProject1/</code>	Paper feed shows empty state
2	Filtering works but shows 0 results	No errors in console

3	Full Summaries section is empty	No crash
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TC-ERR-002: Corrupt Sidecar JSON

Step	Action	Expected Result
1	Place a PDF with an invalid JSON sidecar (syntax error)	Server logs a warning: <code>[warn] sidecar read error</code>
2	Paper still appears in feed	Uses auto-inferred metadata
3	Other papers unaffected	All load correctly

TC-ERR-003: Missing Paper Folder

Step	Action	Expected Result
1	Rename or delete the <code>AI papers for WebProject1/</code> folder	Watcher handles <code>FileNotFoundError</code> gracefully
2	Server keeps running	No crash
3	Restore folder	Watcher picks up PDFs on next poll

TC-ERR-004: Stale Server Process

Step	Action	Expected Result
1	Start server, note PID	—
2	Start server again	Old process killed via PID file; new server starts on port 5000
3	Verify <code>flask.pid</code> updated	Contains new PID

TC-ERR-005: Network Failure — Discovery

Step	Action	Expected Result
1	Disconnect from internet (or block proxy)	—
2	Click "Find New Papers"	Status shows error message (not a crash)
3	Local library features continue working	Feed, filtering, search all functional

TC-ERR-006: Concurrent Paper Upload with Same Name

Step	Action	Expected Result
1	Upload paper .pdf	Succeeds
2	Upload paper .pdf again	Returns 409: "A paper with this filename already exists"

TC-ERR-007: API with Invalid JSON Body

Step	Action	Expected Result
1	PUT /api/papers/{id} with malformed JSON	Handled gracefully (uses <code>get_json(silent=True)</code>)
2	Returns 400 with appropriate error	No server crash

TC-ERR-008: Discovery with ArXiv URL Conversion

Step	Action	Expected Result
1	Submit an arXiv abstract URL (<code>arxiv.org/abs/2401.12345</code>) to discover figure	URL automatically converted to PDF URL (<code>arxiv.org/pdf/2401.12345</code>)
2	Figure extraction proceeds normally	No error from URL format

Appendix A: API Endpoint Reference

Method	Endpoint	Auth	Description
GET/POST	/login	No	Login page and handler
GET/POST	/register	No	Registration page and handler
GET	/logout	Yes	End session
GET	/	Yes	Main page
GET	/styles.css	No	Stylesheet
GET	/app.js	No	Frontend JavaScript
GET	/AI papers for WebProject1/<path>	No	Serve PDFs and sidecars
GET	/Key Word Images/<path>	No	Serve keyword images
GET	/api/papers	Yes	List/search papers
GET	/api/papers/<id>	Yes	Get single paper
POST	/api/papers	Yes	Upload new paper
PUT	/api/papers/<id>	Yes	Update paper metadata
DELETE	/api/papers/<id>	Yes	Delete paper + files
POST	/api/papers/<id>/generate-infographic	Yes	Generate AI infographic
POST	/api/papers/<id>/generate-images	Yes	Generate best-figure + infographic
GET	/api/papers/citation-counts	Yes	Fetch live citation counts
GET	/api/visit-count	Yes	Get visit counter
POST	/api/discover/figure	Yes	Extract figure from discovery PDF
POST	/api/discover/figure/manual-boxes	Yes	Save manual bounding box

POST	/api/discover/pdf-page	Yes	Render a PDF page/region
GET	/api/discover/search	Yes	Search external paper sources
POST	/api/discover/rank	Yes	AI-rank discovery candidates
GET	/api/discover/ranking-criteria	Yes	Get ranking criteria for a team
POST	/api/discover/ranking-criteria	Yes	Update ranking criteria
GET	/api/discover/progress	Yes	Poll discovery progress
POST	/api/clipboard/download-zip	Yes	Bundle PDFs into ZIP
POST	/api/clipboard/report	Yes	Generate LLM report
GET	/api/changes	Yes	SSE live update stream

Appendix B: Test Result Template

Test Case ID	T	D	B	R	Notes
TC-AUTH-001					
TC-AUTH-002					
...					

End of Test Plan